

Directions: Answer each question in the space provided. Close book and closed notes. No electronic devices.

1. (30 points) Short Answer. You do not need to show your work for problems on this page.

- (a) State two different (non-isomorphic) groups of order 9.

$$\mathbb{Z}_9 \text{ and } \mathbb{Z}_3 \times \mathbb{Z}_3$$

- (b) State (in set notation) the cosets of $H = \langle 3 \rangle \subseteq \mathbb{Z}_9$. ($\mathbb{Z}_9$ is abelian, so you may give just the left cosets.)

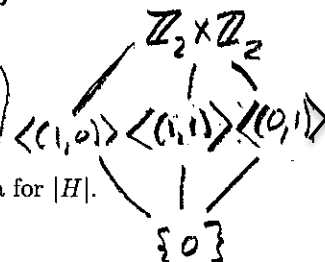
$$H = \{0, 3, 6\}, 1+H = \{1, 4, 7\}, 2+H = \{2, 5, 8\}$$

- (c) Write an addition table for $\mathbb{Z}_2 \times \mathbb{Z}_2$.

+	(0,0)	(1,0)	(0,1)	(1,1)
(0,0)	(0,0)	(1,0)	(0,1)	(1,1)
(1,0)	(1,0)	(0,0)	(1,1)	(0,1)
(0,1)	(0,1)	(1,1)	(0,0)	(1,0)
(1,1)	(1,1)	(0,1)	(1,0)	(0,0)

- (d) Give an example of a finite group that has more subgroups than it has elements.

$\mathbb{Z}_2 \times \mathbb{Z}_2$ has 4 elements and 5 subgroups



- (e) Let g be a generator of a finite cyclic group G , and let $H = \langle g^m \rangle \cap \langle g^n \rangle$. Give a formula for $|H|$.

$$|H| = \frac{|G|}{\gcd(m,n)}$$

- (f) What is the smallest n for which A_n is non-abelian?

$$n=4$$

- (g) Give an example of a permutation $\mu \in S_{12}$ for which $|\mu| = 28$.

$$(1, 2, 3, 4, 5, 6, 7)(8, 9, 10, 11)$$

- (h) Is the permutation μ from part (g) even or odd?

odd (product of even and odd)

- (i) State the order of the permutation $(12376)^{-1}(54) \in S_7$.

$$= (6, 7, 3, 2, 1)(5, 4) \quad \text{order} = \text{lcm}(5, 2) = 10$$

- (j) Write $(1234)^2(35)$ as a product of disjoint cycles.

$$= (13)(24)(35) = (135)(24)$$

2. (10 points) Does S_4 have any subgroups of order 10? If so, then state one. If not, explain why not.

NO $|S_4| = 4! = 24$, and by Lagrange's theorem any subgroup of S_4 has an order that divides 24. Because $10 \nmid 24$, there are no subgroups of order 10.

3. (10 points) Does S_4 have any subgroups of order 4? If so, then state one. If not, explain why not.

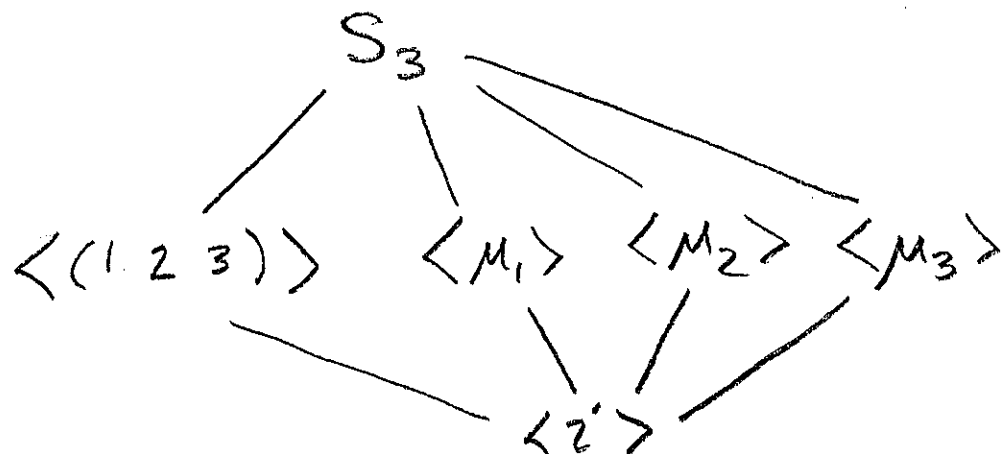
YES Here's one: $H = \langle (1, 2, 3, 4) \rangle$, that is,
 $H = \{ e, (1, 2, 3, 4), (1, 3)(2, 4), (4, 3, 2, 1) \}$

And here's another:

$$K = \{ e, (1, 2), (3, 4), (1, 2)(3, 4) \}$$

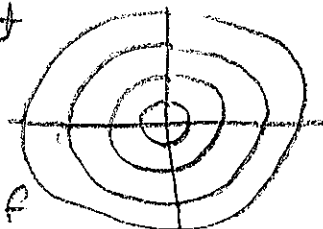
4. (10 points) Prove or disprove: If every proper subgroup of a group G is cyclic, then G is abelian.

This is FALSE. For a counterexample, consider $G = S_3$. It is non-abelian but all its proper subgroups are cyclic:



5. (10 points) Recall that the unit circle $\mathbb{T}$ is a subgroup of $\mathbb{C}^*$. Describe the cosets of $\mathbb{T}$ in $\mathbb{C}^*$. Justify.

Select an arbitrary $z_0 \in \mathbb{C}^*$ and consider $z_0\mathbb{T} = \{z_0 u \mid u \in \mathbb{T}\}$. Any point $z_0 u$ in this set satisfies $|z_0 u| = |z_0| |u| = |z_0| \cdot 1 = |z_0|$, so every point in $z_0\mathbb{T}$ is at a distance of $|z_0|$ from the origin. Conversely, if $u \in \mathbb{C}^*$ is at distance $|z_0|$ from 0, then $u = z_0 (z_0^{-1} u)$ and $|z_0^{-1} u| = |z_0|^{-1} |u| = |z_0|^{-1} |z_0| = 1$, so $u \in z_0\mathbb{T}$. Thus any coset $z_0\mathbb{T}$ is a circle of radius $|z_0|$ centered at 0.



6. (10 points) Suppose g is a fixed element of a group G .

Prove that the set $H = \{x \in G : xg = gx\}$ is a subgroup of G .

Proof Observe that:

① $e \in H$ because $eg = g = ge$.

② H is closed under multiplication:

Suppose $x, y \in H$. This means $xg = gx$ and $yg = gy$. Therefore:

$$\begin{aligned} (xy)g &= x(yg) = x(gy) = (xg)y = (gx)y \\ &= g(xy) \end{aligned}$$

and hence $xy \in H$.

③ H is closed under inverses

Suppose $x \in H$, meaning $xg = gx$. Then

$$\begin{aligned} x^{-1}(xg)x^{-1} &= x^{-1}(gx)x^{-1} \\ gx^{-1} &= x^{-1}g \end{aligned}$$

Consequently $x^{-1} \in H$.

Considerations ①, ②, ③ above show H is a subgroup. $\square$

7. (10 points) Suppose p and q are distinct prime numbers. How many generators does $\mathbb{Z}_{pq}$ have? Explain.

An element $a \in \mathbb{Z}_{pq}$ is not a generator exactly when $\gcd(a, pq) \neq 1$, and this happens whenever a is a multiple of p or q . Thus the following elements of $\mathbb{Z}_{pq}$ are the ones that are not generators:

$$0 \cdot p, 1 \cdot p, 2 \cdot p, 3 \cdot p, \dots, (q-1)p$$

$$1 \cdot q, 2 \cdot q, 3 \cdot q, \dots, (p-1)q$$

Thus $1 + (q-1) + (p-1) = p + q - 1$ elements are not generators, so the number of generators is $pq - (p + q - 1) = pq - p - q + 1 = \boxed{(p-1)(q-1)}$

8. (10 points) Prove: If G is a group and $(ab)^2 = a^2b^2$ for any two $a, b \in G$, then G is abelian.

Proof (Direct) Suppose $(ab)^2 = a^2b^2$ for all $a, b \in G$. Now let x and y be two arbitrary elements of G . Then:

$$(xy)^2 = x^2y^2$$

$$xyxy = xx yy$$

$$x^{-1}(xyxy)y^{-1} = x^{-1}(xx yy)y^{-1}$$

$$eyxe = exye$$

$$yx = xy.$$

Thus $xy = yx$, so G is abelian. $\square$